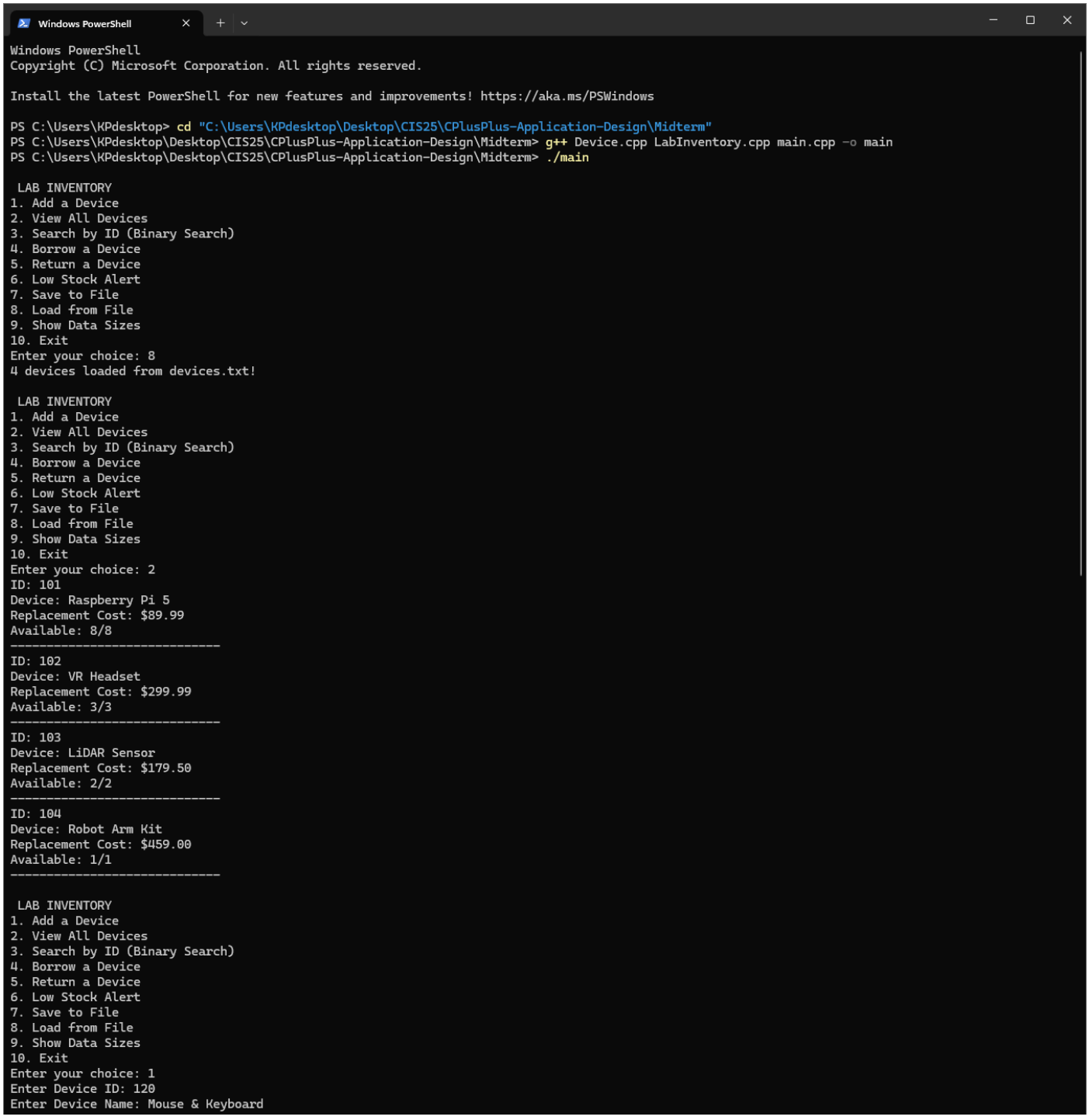
The LAB INVENTORY management system is a C++ program designed to track and manage technology equipment in a lab.

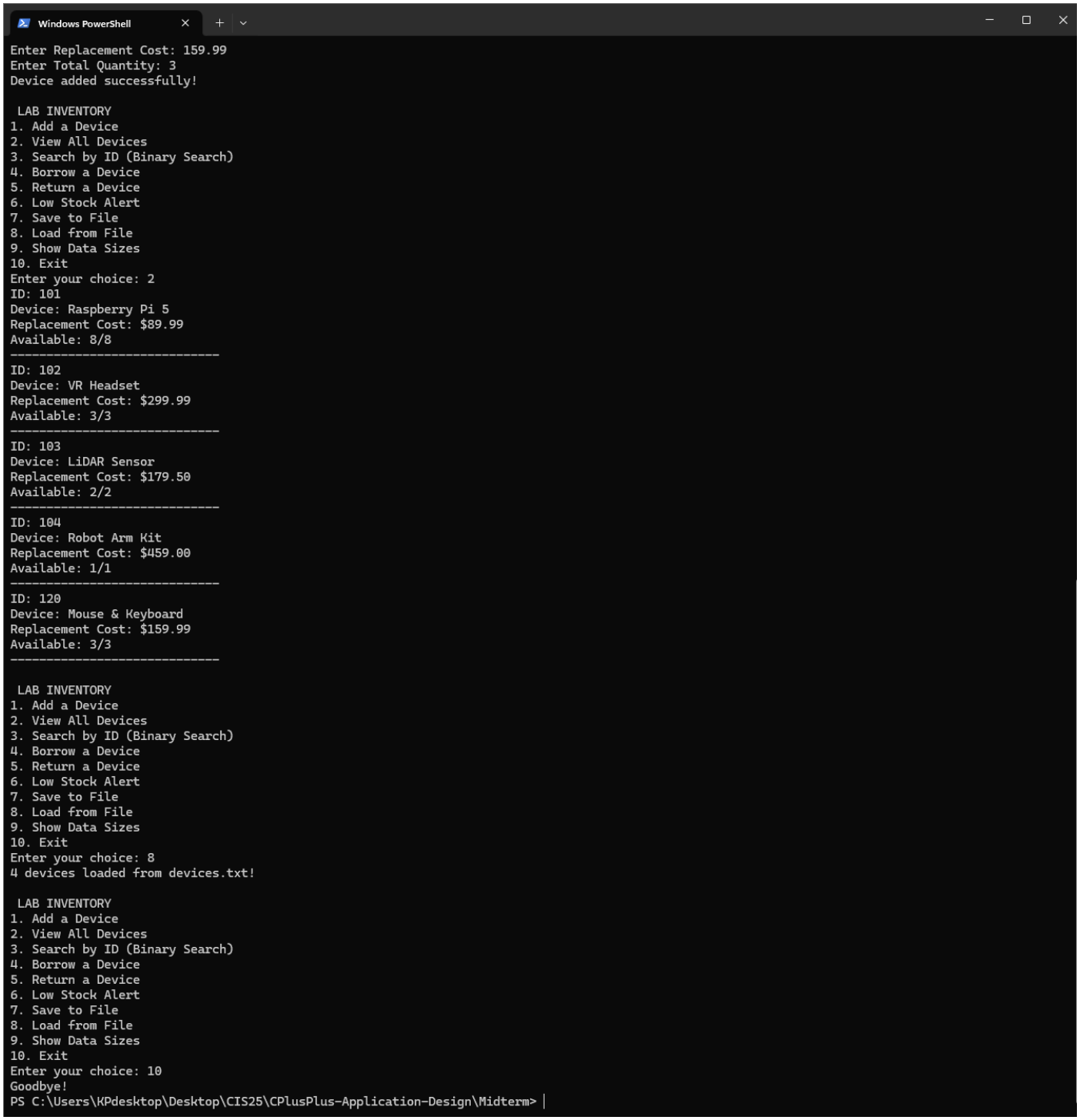
Users can load and save the data from the `device.txt` add and view devices, search for equipment by ID, borrow or return devices, and check low-stock alerts. The program also allows users to save and load inventory data using a text file.

The program uses two classes, `Device` and `LabInventory`, to organize equipment information and inventory operations. It uses arrays and pointers to manage records, while binary search helps users find devices efficiently.

C++ Concepts Used:

1. **Data Types and Sizes:** Uses `int`, `double`, `bool`, and `sizeof()`.
2. **File Types:** Uses a `.txt` file to store inventory records.
3. **Pointers:** Uses pointers and dynamic memory allocation (`new[]` and `delete[]`).
4. **Arrays:** Stores multiple devices in a dynamically allocated array.
5. **Binary Search:** Searches for devices using their IDs.
6. **Strings:** Stores device names using the `string`.
7. **File I/O:** Uses `ifstream` and `ofstream` to load and save data.
8. **Classes:** Uses two classes, `Device` and `LabInventory`


Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\KPdesktop> cd "C:\Users\KPdesktop\Desktop\CIS25\CPlusPlus-Application-Design\Midterm"
PS C:\Users\KPdesktop\Desktop\CIS25\CPlusPlus-Application-Design\Midterm> g++ Device.cpp LabInventory.cpp main.cpp -o main
PS C:\Users\KPdesktop\Desktop\CIS25\CPlusPlus-Application-Design\Midterm> ./main
LAB INVENTORY
1. Add a Device
2. View All Devices
3. Search by ID (Binary Search)
4. Borrow a Device
5. Return a Device
6. Low Stock Alert
7. Save to File
8. Load from File
9. Show Data Sizes
10. Exit
Enter your choice: 8
4 devices loaded from devices.txt!
LAB INVENTORY
1. Add a Device
2. View All Devices
3. Search by ID (Binary Search)
4. Borrow a Device
5. Return a Device
6. Low Stock Alert
7. Save to File
8. Load from File
9. Show Data Sizes
10. Exit
Enter your choice: 2
ID: 101
Device: Raspberry Pi 5
Replacement Cost: $89.99
Available: 8/8
ID: 102
Device: VR Headset
Replacement Cost: $299.99
Available: 3/3
ID: 103
Device: LiDAR Sensor
Replacement Cost: $179.50
Available: 2/2
ID: 104
Device: Robot Arm Kit
Replacement Cost: $459.00
Available: 1/1
LAB INVENTORY
1. Add a Device
2. View All Devices
3. Search by ID (Binary Search)
4. Borrow a Device
5. Return a Device
6. Low Stock Alert
7. Save to File
8. Load from File
9. Show Data Sizes
10. Exit
Enter your choice: 1
Enter Device ID: 120
Enter Device Name: Mouse & Keyboard


Windows PowerShell
Enter Replacement Cost: 159.99
Enter Total Quantity: 3
Device added successfully!
LAB INVENTORY
1. Add a Device
2. View All Devices
3. Search by ID (Binary Search)
4. Borrow a Device
5. Return a Device
6. Low Stock Alert
7. Save to File
8. Load from File
9. Show Data Sizes
10. Exit
Enter your choice: 2
ID: 101
Device: Raspberry Pi 5
Replacement Cost: $89.99
Available: 8/8
ID: 102
Device: VR Headset
Replacement Cost: $299.99
Available: 3/3
ID: 103
Device: LiDAR Sensor
Replacement Cost: $179.50
Available: 2/2
ID: 104
Device: Robot Arm Kit
Replacement Cost: $459.00
Available: 1/1
ID: 120
Device: Mouse & Keyboard
Replacement Cost: $159.99
Available: 3/3
LAB INVENTORY
1. Add a Device
2. View All Devices
3. Search by ID (Binary Search)
4. Borrow a Device
5. Return a Device
6. Low Stock Alert
7. Save to File
8. Load from File
9. Show Data Sizes
10. Exit
Enter your choice: 8
4 devices loaded from devices.txt!
LAB INVENTORY
1. Add a Device
2. View All Devices
3. Search by ID (Binary Search)
4. Borrow a Device
5. Return a Device
6. Low Stock Alert
7. Save to File
8. Load from File
9. Show Data Sizes
10. Exit
Enter your choice: 10
Goodbye!
PS C:\Users\KPdesktop\Desktop\CIS25\CPlusPlus-Application-Design\Midterm>